1.05 Trigonometry				
1.05a 1.05d	sin, cos and tan for all arguments	a) Understand and be able to use the definitions of sine, cosine and tangent for all arguments.	d) Be able to work with radian measure, including use for arc length and area of sector.	ME1
1.05b	Sine and cosine rules	b) Understand and be able to use the sine and cosine rules.	*Learners should know the formulae $s = r\theta$ and $A = \frac{1}{2}r^2\theta$.*	
	Radians	*Questions may include the use of bearings and require the use of the ambiguous case of the sine rule.*	*Learners should be able to use the relationship between degrees and radians.*	
1.05c		c) Understand and be able to use the area of a triangle in the form $\frac{1}{2}ab\sin C$.		
1.05e	Small angle approximations		e) Understand and be able to use the standard small angle approximations of sine, cosine and tangent: 1. $\sin\theta \approx \theta$, 2. $\cos\theta \approx 1 - \frac{1}{2}\theta^2$, 3. $\tan\theta \approx \theta$, where θ is in radians. *e.g. Find an approximate expression for* $\dfrac{\sin 3\theta}{1+\cos\theta}$ *if θ is small enough to neglect terms in θ^3 or above.*	ME2
1.05f 1.05g	Graphs of the basic trigonometric functions Exact values of trigonometric functions	f) Understand and be able to use the sine, cosine and tangent functions, their graphs, symmetries and periodicities. *Includes knowing and being able to use exact values of* $\sin\theta$ *and* $\cos\theta$ *for* $\theta = 0°, 30°, 45°, 60°, 90°, 180°$ *and multiples thereof and exact values of* $\tan\theta$ *for* $\theta = 0°, 30°, 45°, 60°, 180°$ *and multiples thereof.*	g) Know and be able to use exact values of $\sin\theta$ and $\cos\theta$ for $\theta = 0, \frac{1}{6}\pi, \frac{1}{4}\pi, \frac{1}{3}\pi, \frac{1}{2}\pi, \pi$ and multiples thereof, and exact values of $\tan\theta$ for $\theta = 0, \frac{1}{6}\pi, \frac{1}{4}\pi, \frac{1}{3}\pi, \pi$ and multiples thereof.	ME3

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.05h 1.05i	Inverse and reciprocal trigonometric ratios		h) Understand and be able to use the definitions of secant ($\sec\theta$), cosecant ($\operatorname{cosec}\theta$) and cotangent ($\cot\theta$) and of $\arcsin\theta$, $\arccos\theta$ and $\arctan\theta$ and their relationships to $\sin\theta$, $\cos\theta$ and $\tan\theta$ respectively. i) Understand the graphs of the functions given in 1.05h, their ranges and domains. *In particular, learners should know that the principal values of the inverse trigonometric relations may be denoted by* $\arcsin\theta$ *or* $\sin^{-1}\theta$, $\arccos\theta$ *or* $\cos^{-1}\theta$, $\arctan\theta$ *or* $\tan^{-1}\theta$ *and relate their graphs (for the appropriate domain) to the graphs of* $\sin\theta$, $\cos\theta$ *and* $\tan\theta$.	ME4
1.05j 1.05k	Trigonometric identities	j) Understand and be able to use $\tan\theta \equiv \frac{\sin\theta}{\cos\theta}$ and $\sin^2\theta + \cos^2\theta \equiv 1$. *In particular, these identities may be used in solving trigonometric equations and simple trigonometric proofs.*	k) Understand and be able to use $\sec^2\theta \equiv 1 + \tan^2\theta$ and $\operatorname{cosec}^2\theta \equiv 1 + \cot^2\theta$. *In particular, the identities in 1.05j and 1.05k may be used in solving trigonometric equations, proving trigonometric identities or in evaluating integrals.*	ME5

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.05l 1.05m 1.05n	Further trigonometric identities		l) Understand and be able to use double angle formulae and the formulae for $\sin(A \pm B)$, $\cos(A \pm B)$ and $\tan(A \pm B)$. *Learners may be required to use the formulae to prove trigonometric identities, simplify expressions, evaluate expressions exactly, solve trigonometric equations or find derivatives and integrals.* m) Understand the geometrical proofs of these formulae. n) Understand and be able to use expressions for $a\cos\theta + b\sin\theta$ in the equivalent forms of $R\cos(\theta \pm \alpha)$ or $R\sin(\theta \pm \alpha)$. *In particular, learners should be able to:* *1. sketch graphs of* $a\cos\theta + b\sin\theta$*,* *2. determine features of the graphs including minimum or maximum points and* *3. solve equations of the form* $a\cos\theta + b\sin\theta = c$.	ME6
1.05o	Trigonometric equations	o) Be able to solve simple trigonometric equations in a given interval, including quadratic equations in $\sin\theta$, $\cos\theta$ and $\tan\theta$ and equations involving multiples of the unknown angle. *e.g.* $\sin\theta = 0.5$ for $0 \le \theta < 360°$ $6\sin^2\theta + \cos\theta - 4 = 0$ for $0 \le \theta < 360°$ $\tan 3\theta = -1$ for $-180° < \theta < 180°$	*Extend their knowledge of trigonometric equations to include radians and the trigonometric identities in Stage 2.*	ME7

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.05p	Proof involving trigonometric functions		p) Be able to construct proofs involving trigonometric functions and identities. *e.g. Prove that* $\cos^2\left(\theta + 45^\circ\right) - \frac{1}{2}(\cos 2\theta - \sin 2\theta) = \sin^2\theta$. *Includes constructing a mathematical argument as described in Section 1.01.*	ME8
1.05q	Trigonometric functions in context		q) Be able to use trigonometric functions to solve problems in context, including problems involving vectors, kinematics and forces. *Problems may include realistic contexts, e.g. movement of tides, sound waves, etc. as well as problems in vector form which involve resolving directions and quantities in mechanics.*	ME9